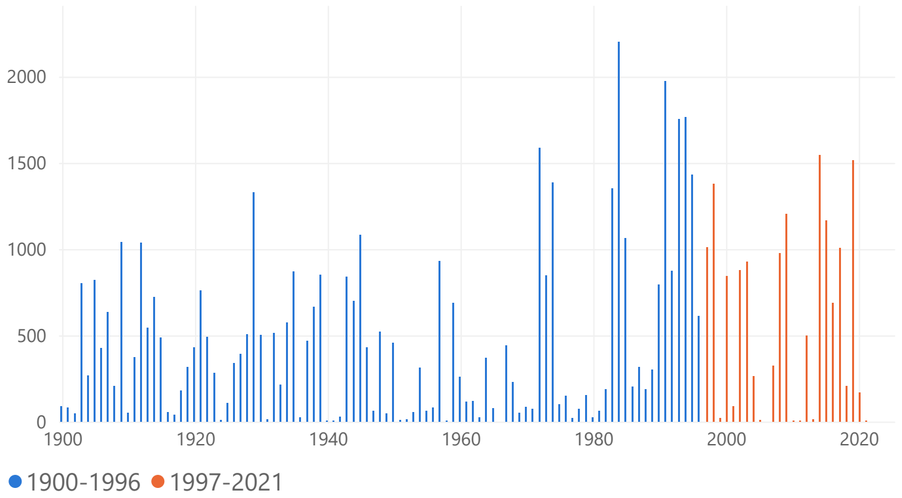
Industry Iceberg Exposure

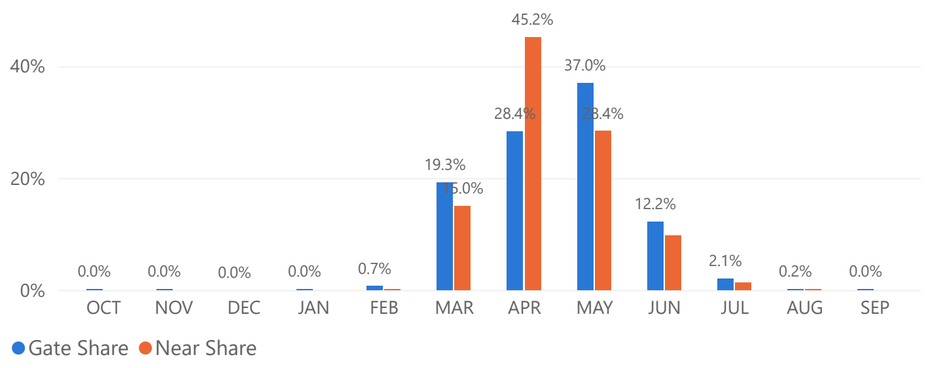
Icebergs Drifting South Past 48°N, 1900–2021

by International Ice Patrol, blue = before production, orange = production era

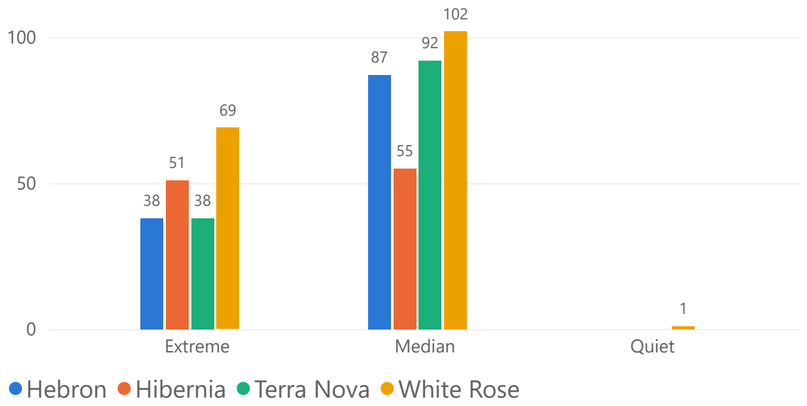


Period	Gate Min	Min Year	Gate Median	Median Year	Gate Max	Max Year	Years Under 50
1900-1996	0	1966	317	1919	2202	1984	16
1997-2021	0	2006	499	2012	1546	2014	7

Share of season: gate count (1997–2021) vs unique bergs at closest approach within 50 km



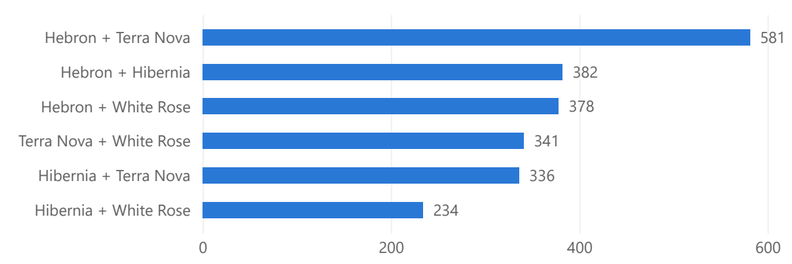
The median year brought more icebergs near the platforms than the extreme year



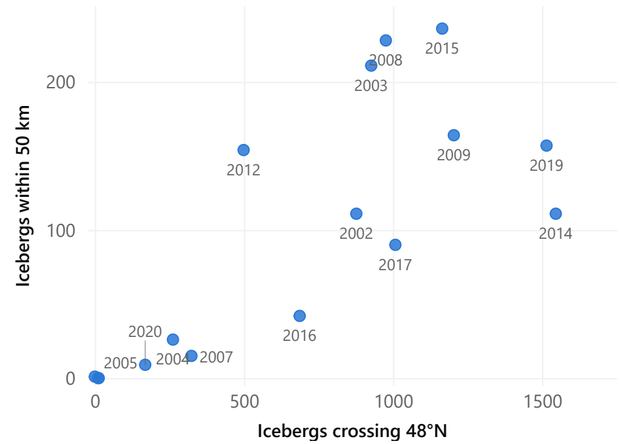
Unique icebergs within 50 km by size (size at closest approach)

Year	Hebron	Hibernia	Terra Nova	White Rose	Total
Extreme	38	51	38	69	**111**
Median	87	55	92	102	**154**
Quiet				1	**1**
Total	**125**	**106**	**130**	**172**	**266**

Unique bergs within 50 km of both facilities on the same day



Regional Count vs Count Near the Platforms



Distance between facilities (km)

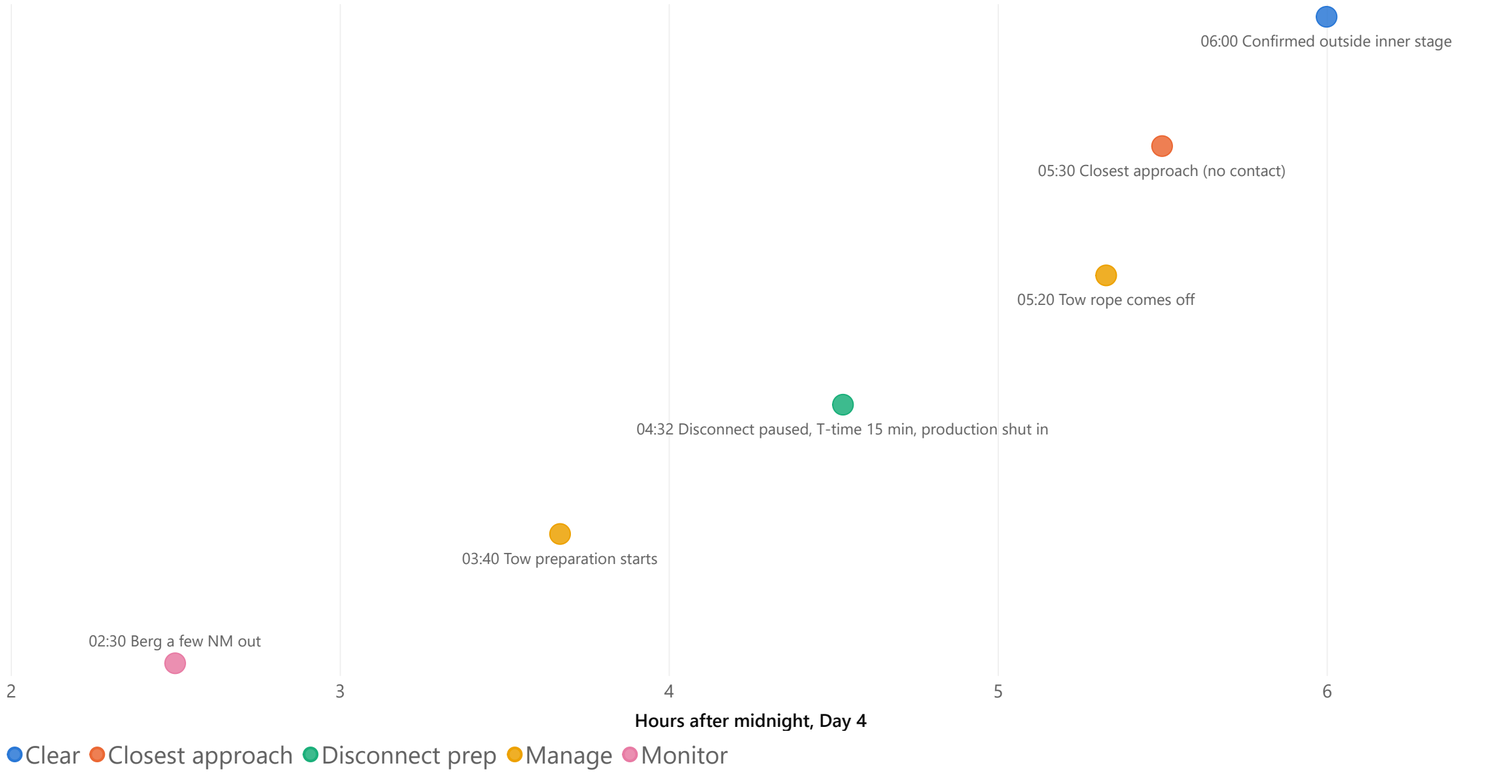
Facility	Hebron	Hibernia	Terra Nova	White Rose
Hebron	0.00	31.60	8.20	46.10
Hibernia	31.60	0.00	38.90	59.10
Terra Nova	8.20	38.90	0.00	49.90
White Rose	46.10	59.10	49.90	0.00

- *More icebergs regionally ≠ more near the platforms (2012 > 2014)*
- *April = peak risk near the platforms*
- *Hebron & Terra Nova: 8.2 km apart, same icebergs, different options. Only Terra Nova can leave.*

Responding to An Iceberg

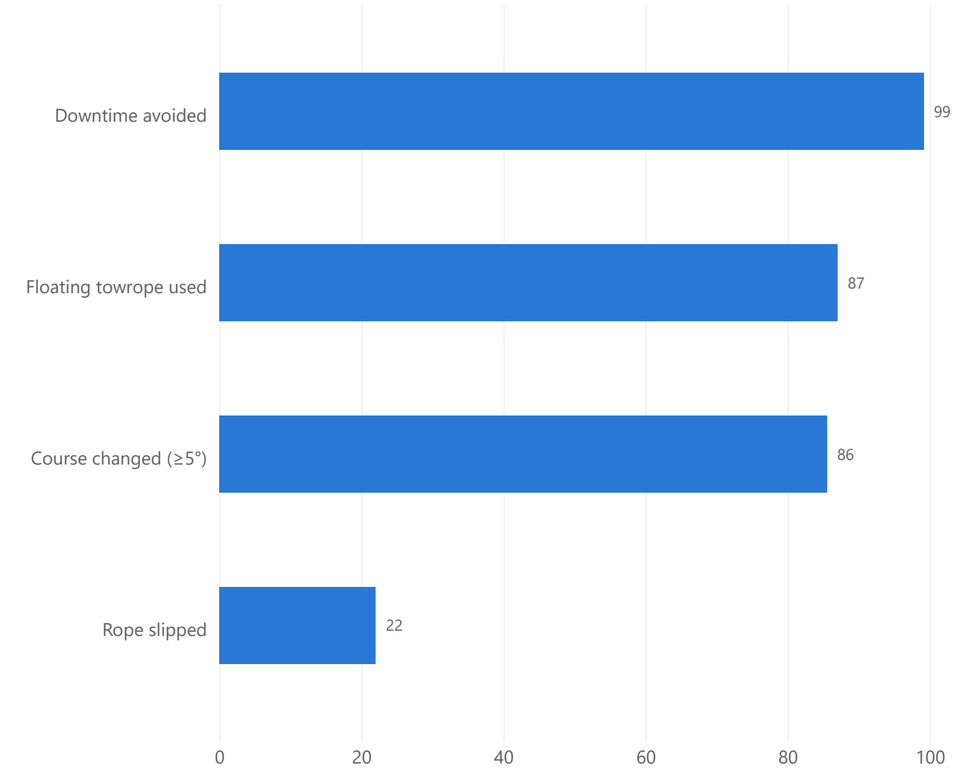
Example Iceberg Response: Final Three Hours

Illustrative timeline from the case pack · Days 1–3: berg spotted by recon flight, then watched by a standby vessel



Tows mostly work, but 1 in 5 ropes slipped

% of past tow operations (Rudkin et al. 2005) · a tow's outcome is unknown until …



WHO DECIDES · offshore vs onshore

Offshore: the platform manager (OIM) alone decides to disconnect

Onshore: supports with data, people and a second opinion

WHO PROVIDES WHAT · one platform vs all four operators

Each platform: its ice plan, alert zones and disconnect decision

All four together: one shared iceberg map, flights and a vessel pool

Recommendation: never wait on a tow, start disconnect preparation as soon as an iceberg enters the middle alert zone

Iceberg Detection

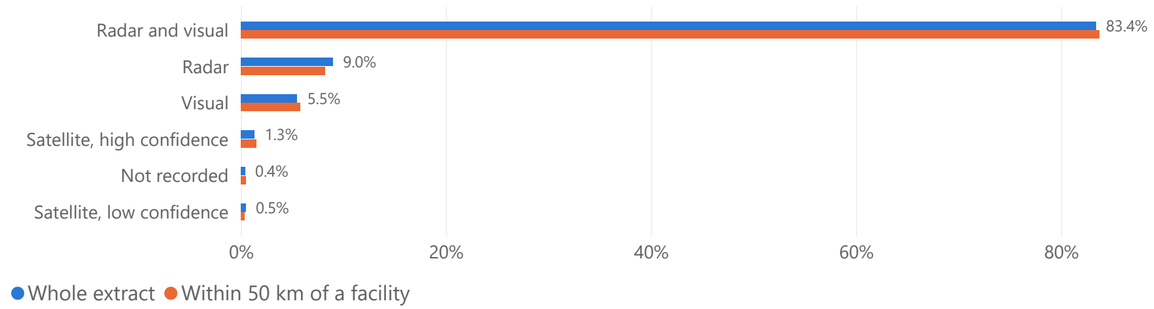
Icebergs within 50 km
1,555

Median hours to closest
38.3 h

First seen at closest
40.6%

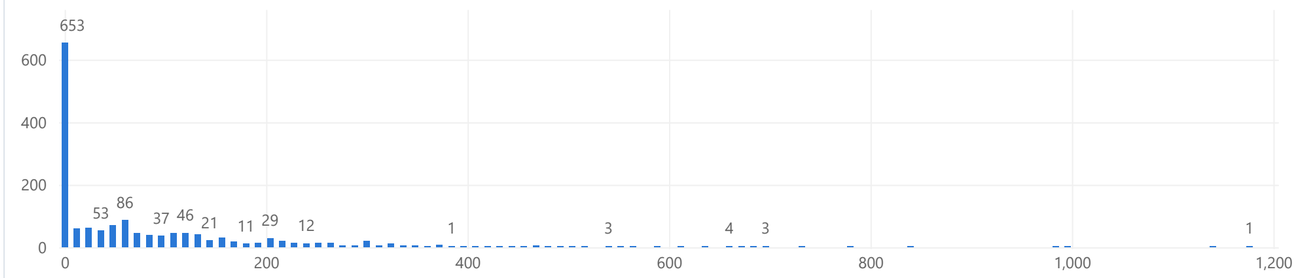
Detection Method Mix

Share of sightings



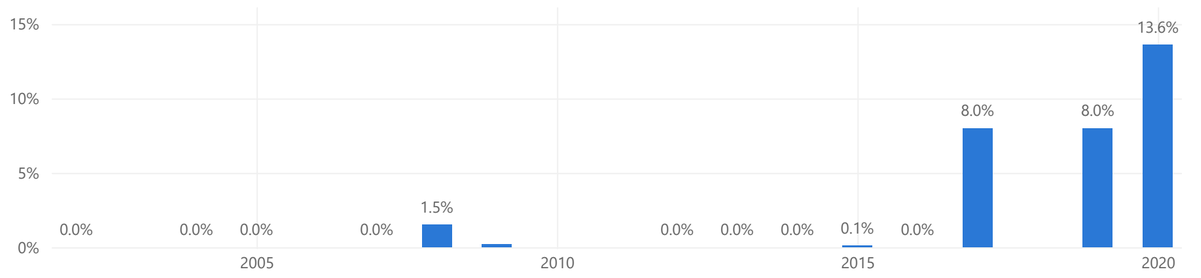
Time from First Record to Closest Approach

Icebergs within 50 km · first record from the dataset· 12 h bins · not a warning time



Satellite Share of Sightings by Year

Share of sighting rows from satellite · in complete dataset



KEEP IN-HOUSE · each platform
Staff its own 24/7 iceberg watch
Set its own alert zones
Retain sole authority to disconnect

OUTSOURCE · seasonal contracts
Spotter flights
Towing vessels
Satellite imagery

POOL ACROSS OPERATORS · working group
One shared iceberg tracking system
Jointly procured flights and satellite
Shared vessel fleet for heavy years

Keep watching far out: 41% of icebergs near the platforms weren't seen until their closest point. Don't judge satellites by past use (under 2% of sightings).

Production Consequences

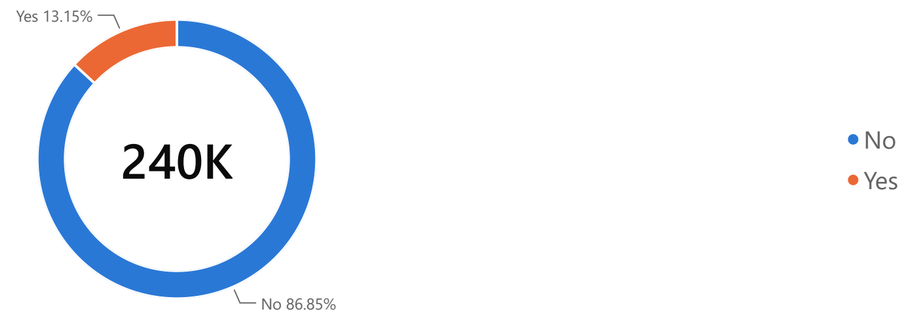
Daily Production and Value by Field

2025 barrels per day · Brent 2025 $69.14/bbl · price × barrels, not profitable

Field	System	Can disconnect	Barrels Per Day	One Day's Value (USD)
Hebron	Gravity-based structure	No	137,714	$9,521,546
Hibernia	Gravity-based structure	No	70,711	$4,888,959
North Amethyst	Subsea satellite	Yes	2,435	$168,356
Terra Nova	FPSO	Yes	22,050	$1,524,537
White Rose	FPSO	Yes	7,074	$489,096
Total			**239,984**	**$16,592,494**

Output That Can Disconnect

Share of 2025 barrels per day · FPSO-linked fields vs fixed platforms



Monthly Offshore Production, 2014

Total barrels per day · extreme iceberg year · spring = May - April

Month	Barrel/day	season
January	227,846.45	Rest of year
February	242,168.71	Rest of year
March	243,212.32	Spring (Mar-Jun)
April	203,757.20	Spring (Mar-Jun)
May	233,385.81	Spring (Mar-Jun)
June	222,682.40	Spring (Mar-Jun)
July	213,353.06	Rest of year
August	153,561.97	Rest of year
September	208,532.53	Rest of year
October	181,382.71	Rest of year
November	245,634.50	Rest of year
December	219,900.26	Rest of year
Total	**2,595,417.93**	

Monthly Iceberg Arrivals, 2014

Icebergs crossing 48°N that month, 2014

Month	Sum of gate_count
January	0
February	15
March	268
April	541
May	532
June	172
July	16
August	2
September	0
October	0
November	0
December	0
Total	**1546**

Production Stopped per Day Off Station

2025 rates · days off station are not in the pack

Unit Leaving Station	Fields stopped	Barrels/day stopped	Value per day (USD)
Both FPSOs	All three	31559	2181989
SeaRose FPSO	White Rose + North Amethyst	9509	657452
Terra Nova FPSO	Terra Nova	22050	1524537

FPSOs can leave a berg's path at $0.66–1.52M a day in lost output; Hebron and Hibernia can't move, so $14.4M a day depends on early detection · spring 2014 output didn't drop

Operating Plan: Quiet Year vs Extreme Year

QUIET SEASON SIGNAL
Under 10 icebergs at 48°N by end of March
Past 9 such years: never more than 330 all season

UNCERTAIN SEASON SIGNAL
10–199 icebergs at 48°N by end of March
Past 9 such years ended anywhere from 89 to 1,380
→ Normal setup, extra ships and flights on standby

EXTREME SEASON SIGNAL
200+ icebergs at 48°N by end of March
Past 7 such years: every one ended with 687 or more
→ Switch to the extreme-year plan from April

OPERATORS · QUIET-YEAR POSTURE
Check the ice plan before 1 March
Normal radar, lookout and standby vessel
One disconnect practice for FPSOs

OPERATORS · EXTREME-YEAR POSTURE
Practise disconnect in March, before the peak
One person checks alert stages every shift
Start disconnect prep early, even during a tow

Prep disconnect early, don't wait on a tow

WORKING GROUP · EXTREME-YEAR POSTURE
Book extra flights and satellite together
Share towing ships, fixed platforms first
Alert Hebron and Terra Nova together (8.2 km apart)

WORKING GROUP · QUIET-YEAR POSTURE
Keep one shared iceberg map running
Baseline spotter flights only
One shared standby ship for all four

Be ready by 1 March · count icebergs at end of March to choose quiet or extreme mode · share the map and ships, but each platform makes its own disconnect call